

Basic Principles of Medicine II

Module: Endocrine and Reproductive Systems

DLA

Teratogenesis

St Georges University ©

Assigned Reading

Moore KL, Persaud TVN, Torchia MG. The Developing Human – Clinically Oriented Embryology, 11th Ed. pgs. 433 – 437, 456 – 458

•Pay special attention to the Green boxes for the clinical correlations for this topic.

Objectives

- SOM.1ai.BPM2.2.ER.1.ANAT.RP.1617 Describe the developmental anomalies induced after infection by rubella virus, varicella virus, cytomegalovirus, toxoplasma gondii, treponema pallidum, herpes simplex
- SOM.1ai.BPM2.2.ER.1.ANAT.RP.1616 Describe the congenital defects associated with maternal diabetes.
- SOM.1ai.BPM2.2.ER.1.ANAT.RP.1613 Describe the mechanism and clinical presentation for congenital anomalies associated with in utero exposure to the following: cigarette smoke, alcohol ingestion, tetracyclines, retinoic acid, warfarin, phenytoin, valproic acid, cocaine, organic mercury
- SOM.1ai.BPM2.2.ER.1.ANAT.RP.1614 Describe the developmental anomalies associated with thalidomide.
- SOM.1ai.BPM2.2.ER.1.ANAT.RP.1615 List and identify the locations where expression of fibroblast growth factor is diminished after exposure to thalidomide and those associated with phocomelia.

Birth Defects



Teros = monster
Teratology –Study of birth defects

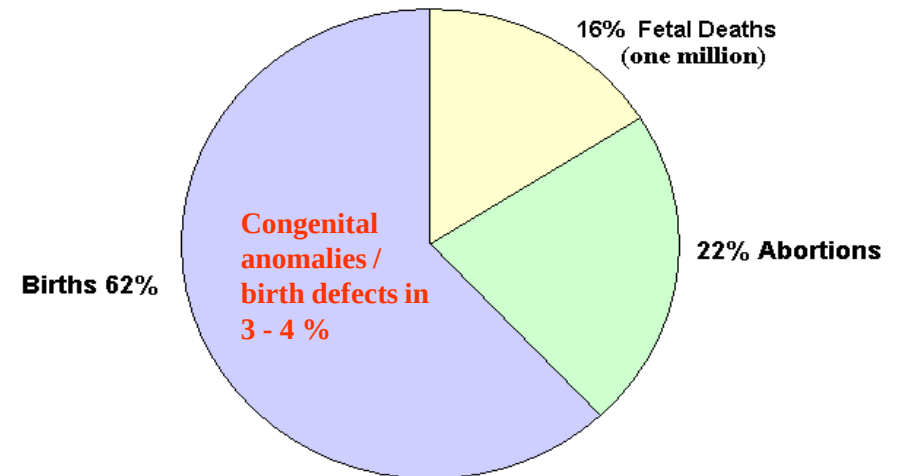




<https://en.wikipedia.org>

What are birth defects?

- Structural or functional anomalies that occur during intrauterine life and can be identified prenatally, at birth or later in life
- Clinically significant birth defects
 - Malformation
 - Disruption
 - Deformation
 - Dysplasia



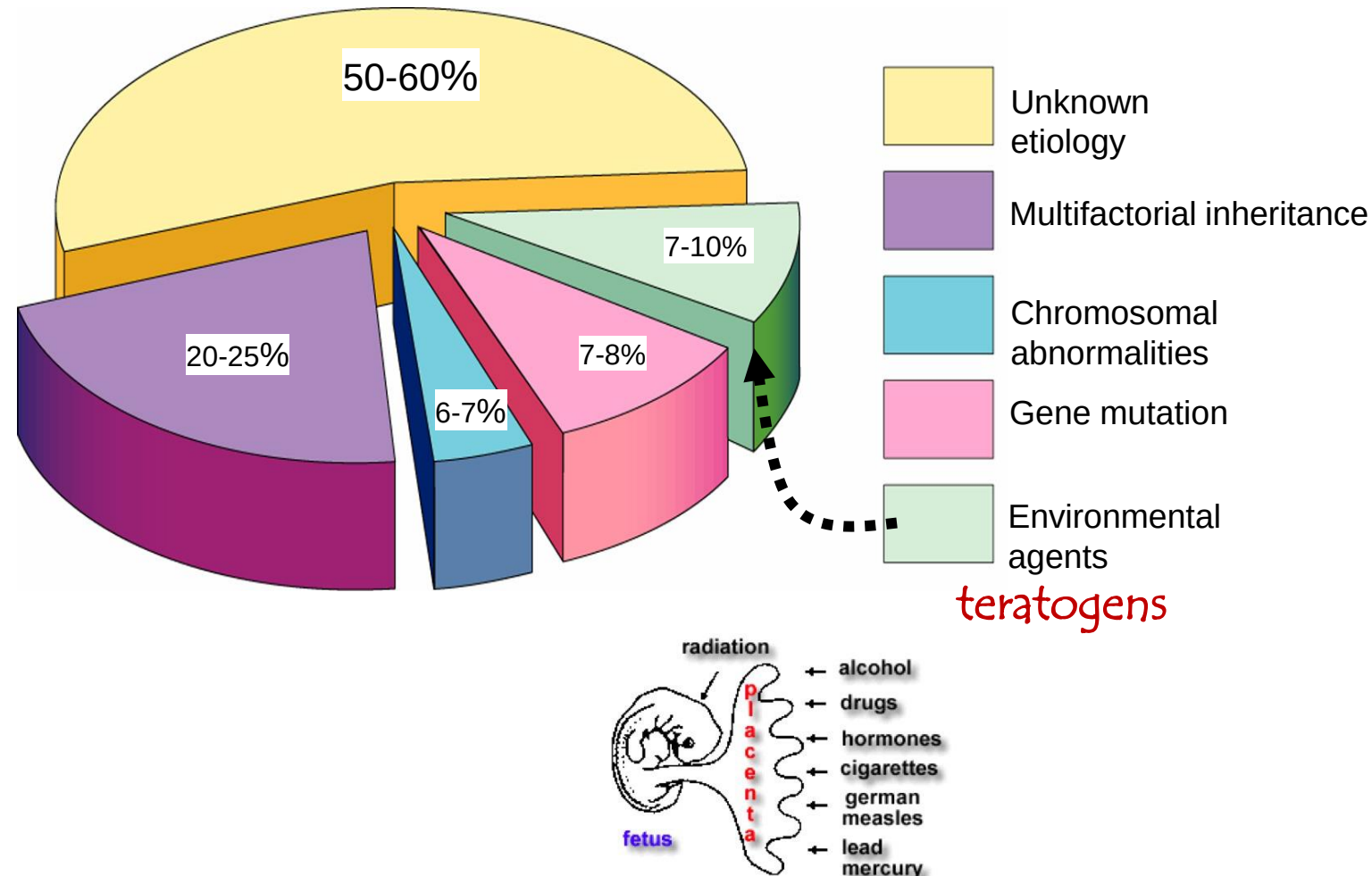
Etiology of human birth defects

Teratogen: An agent which produces a permanent structural or functional alteration in an organism exposed during embryonic or fetal life

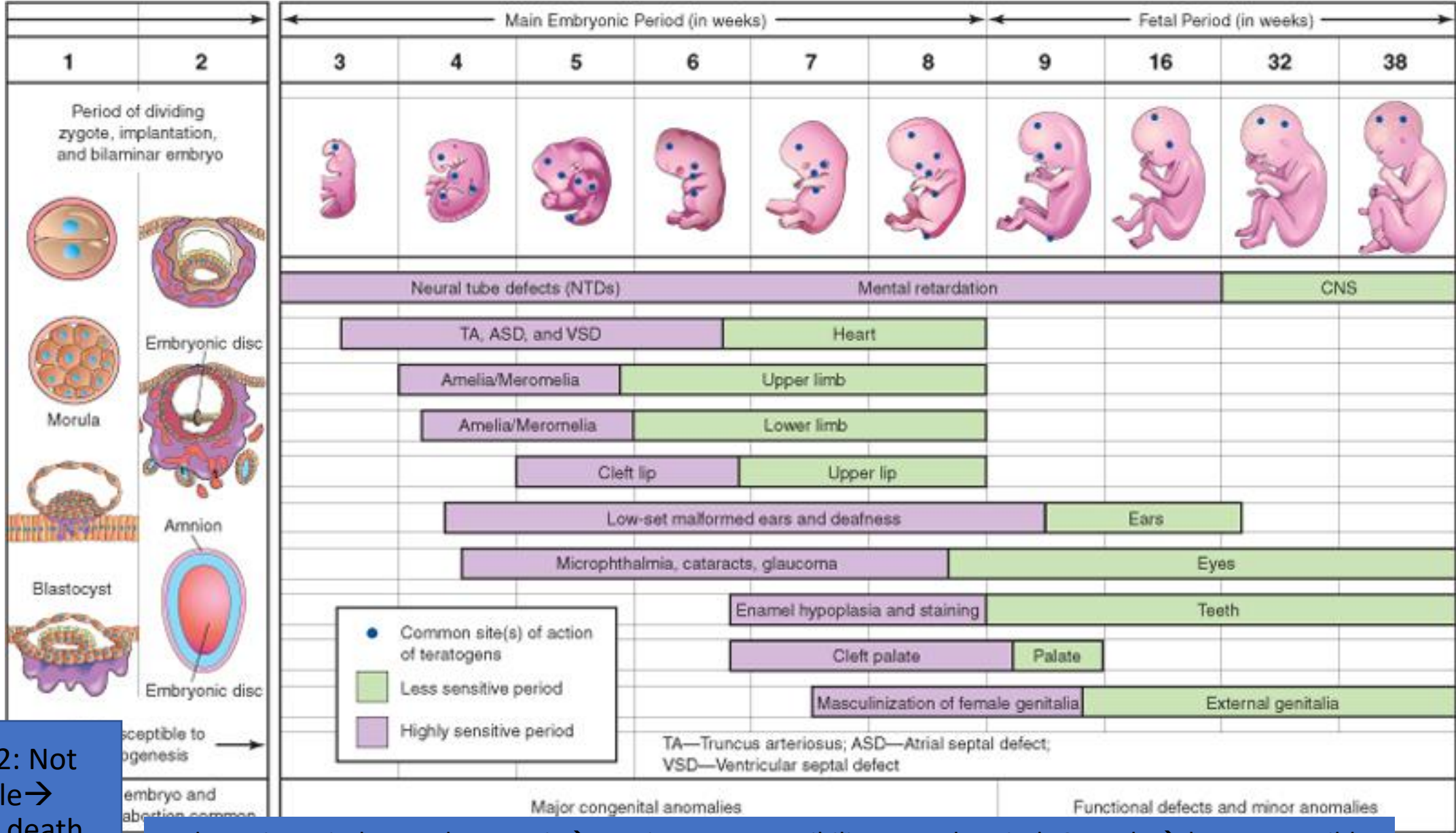
Teratogenesis: The process by which birth defects are produced

Principles:

- Dose of teratogen
- Genotype
- Time of action of the agent: increased teratogenicity at **critical** period of development



Critical Periods in prenatal development



Weeks 1-2: Not susceptible → embryonic death

Embryonic period: Morphogenesis → maximum susceptibility Fetal period: Growth → less susceptible

Clinically important Teratogens

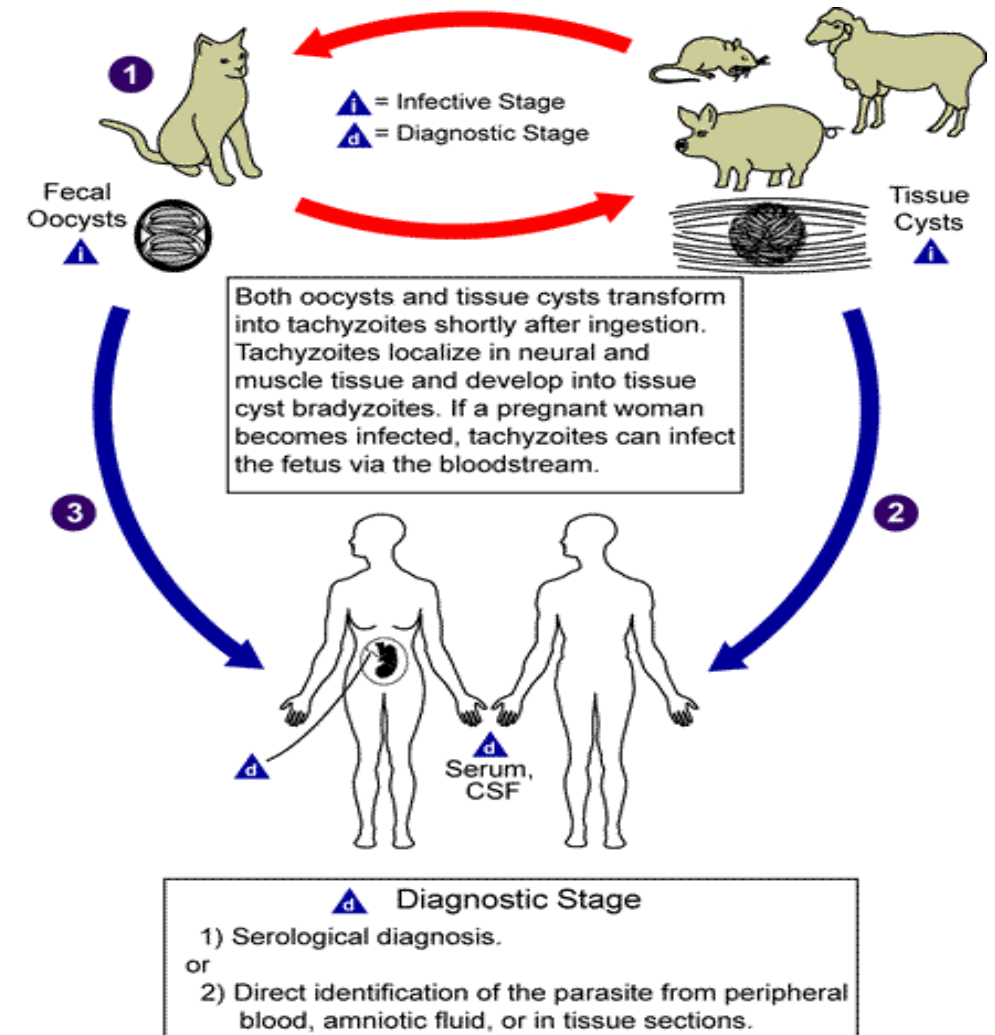
- Physical:
 - Ionizing radiation
- Maternal diseases as teratogens:
 - Uncontrolled Diabetes
- Drugs/medication:
 - Thalidomide
 - Valproic acid
 - Phenytoin
 - Warfarin
 - Tetracycline
 - Isotretinoin
 - Alcohol
- Illicit substances:
 - Cocaine
 - Marijuana
- Chemicals: Methylmercury
- Infections:
 - Toxoplasma Gondii (Toxoplasmosis)
 - Other (Syphilis, varicella-zoster, parvovirus B19)
 - Rubella virus (German measles)
 - Cytomegalovirus (CMV)
 - Herpes simplex virus

Infections as Teratogens

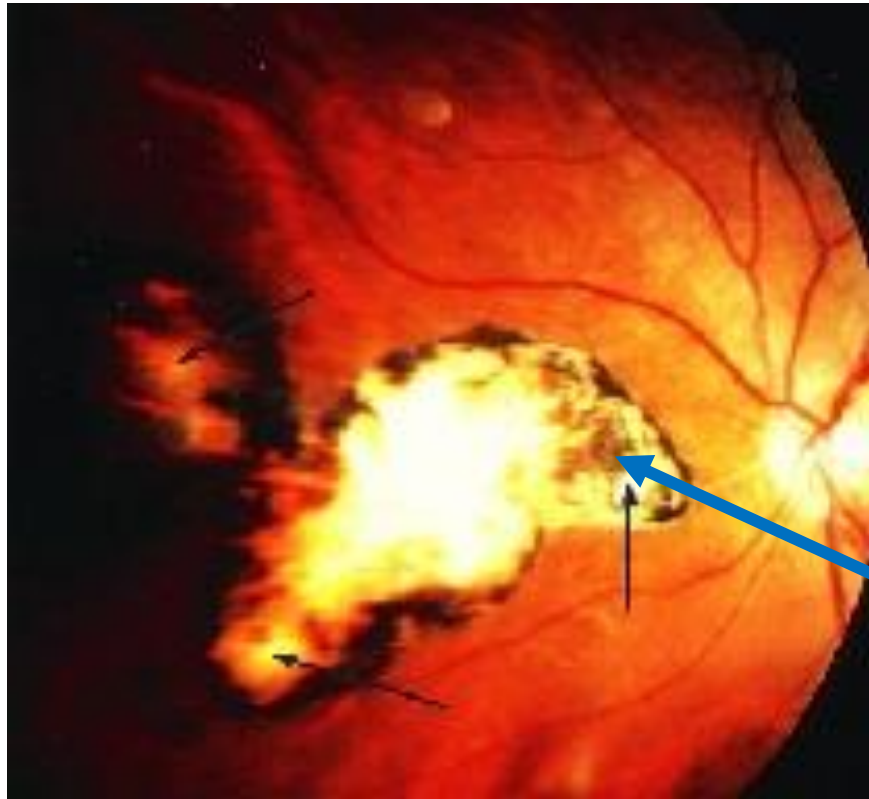
- **T**oxoplasma Gondii (Toxoplasmosis)
- **O**ther (**Syphilis**, **varicella-zoster**, parvovirus B19)
- **R**ubella virus (German measles)
- **C**ytomegalovirus (CMV)
- **H**erpes simplex virus

Congenital Toxoplasmosis

- Caused by protozoan *Toxoplasma Gondii* an intracellular parasite.
- Prognosis:
 - severity related to gestational age and time of infection.
- Management:
 - Antenatal screening
 - Treatment during pregnancy reduces the risk of transmission.

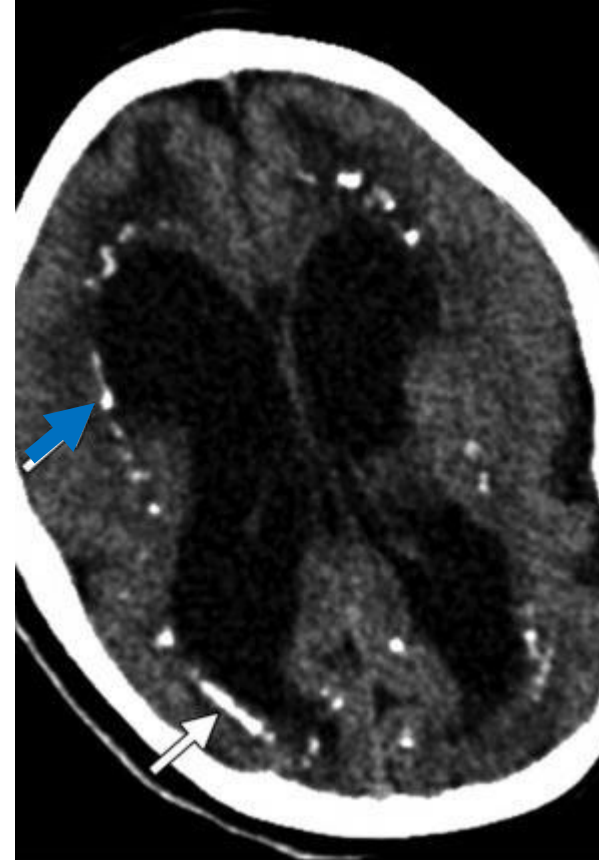


Congenital Toxoplasmosis



Intracranial
calcifications

Chorioretinitis



<https://pedclerk.bsd.uchicago.edu>

At birth most children are **asymptomatic** but may develop symptoms later.

Classic signs: Chorioretinitis, hydrocephalus, intracranial calcifications, convulsions.

Congenital Syphilis

- Syphilis is caused by the bacteria *Treponema pallidum*
- Transmission:
 - Sexually transmitted
 - Transplacental → may occur at any stage of pregnancy
- Clinical manifestations:
 - Stillbirths (hydrops fetalis),
 - Congenital syphilis: deafness, hydrocephalus, intellectual disability, craniofacial anomalies
 - Early or late presentation of symptoms
- Management
 - Prevention : antenatal serological screening
 - Treatment with penicillin
- Incidence increasing due to association with HIV infection



Rash of early congenital syphilis . Hydrocephalus, meningitis, bloody nasal secretions and lymphadenopathy

Late congenital syphilis



Saddle nose



Saber shins

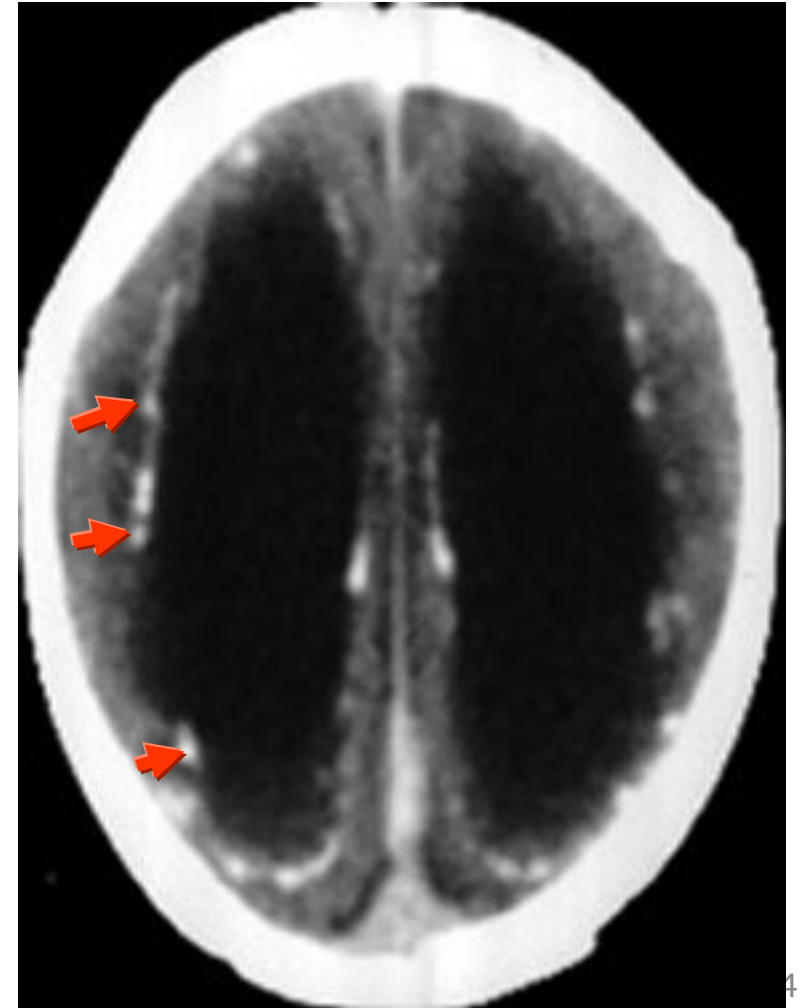


Hutchinson teeth

Congenital Cytomegalovirus (CMV)

- One of the most common viral infections in the fetus.
- Induces abortions if the mother is infected during the early development
- 90% children are asymptomatic at birth
- Up to 10-15% develop symptoms later, notably sensorineural hearing loss
- Other defects : microcephaly, hydrocephalus, microphthalmia, intellectual disability

Ventriculomegaly and
periventricular
calcifications



Herpes Simplex in Pregnancy

- Virus that causes cold sores and genital warts
- Abortions in early pregnancy
- In late pregnancy premature birth and malformations can occur
- Fetal infections most frequently occur during delivery
 - **Mucocutaneous**: Eye, mouth and skin eruptions within 48 hours
 - **CNS**: microcephaly, chorioretinitis and mental deficiency
 - **Disseminated**
- Management:
 - Early diagnosis in mother
 - Antiviral therapy



<http://www.merckmanuals.com>



<https://www.intechopen.com>

Congenital Rubella Syndrome

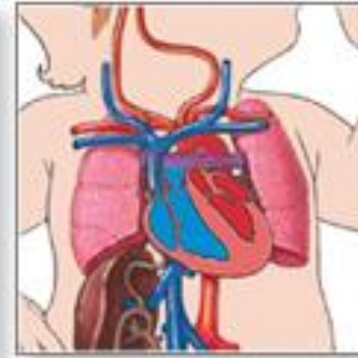
- “German measles “ Rubella virus infection

Features:

- **Cataract** and retinopathy
- **Cardiopathies**: PDA, pulmonary artery stenosis
- CNS : microcephaly and mental deficiency
- **Deafness**
- Skin lesions
- **Prophylaxis with vaccine (MMR)**



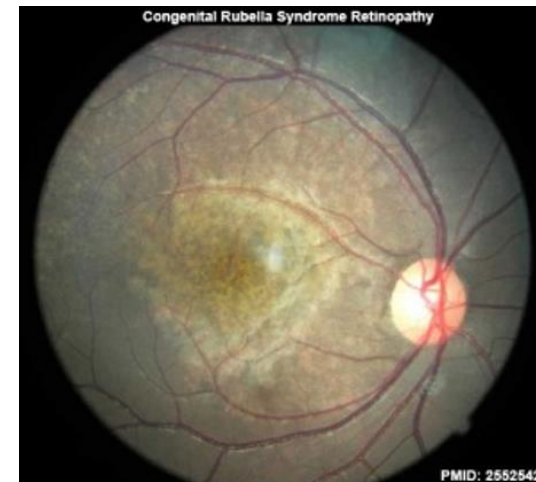
Microcephaly



PDA



Cataracts



<https://embryology.med.unsw.edu.au>

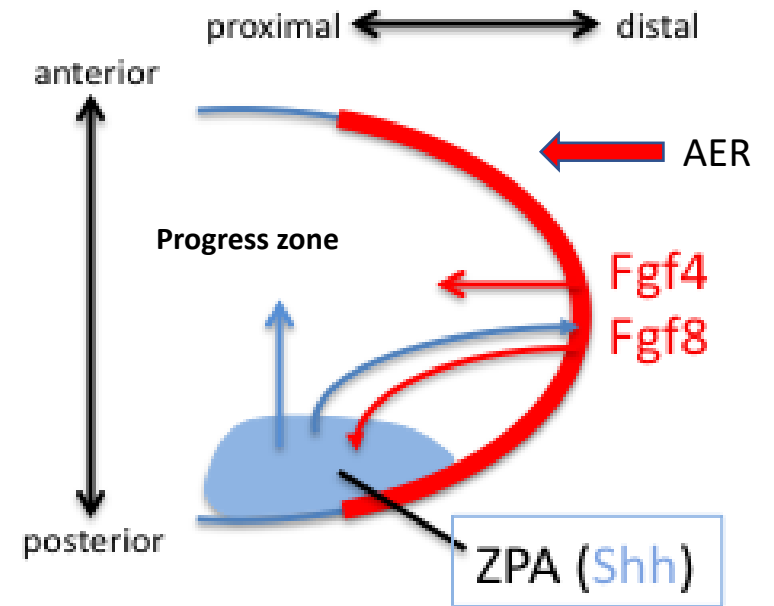
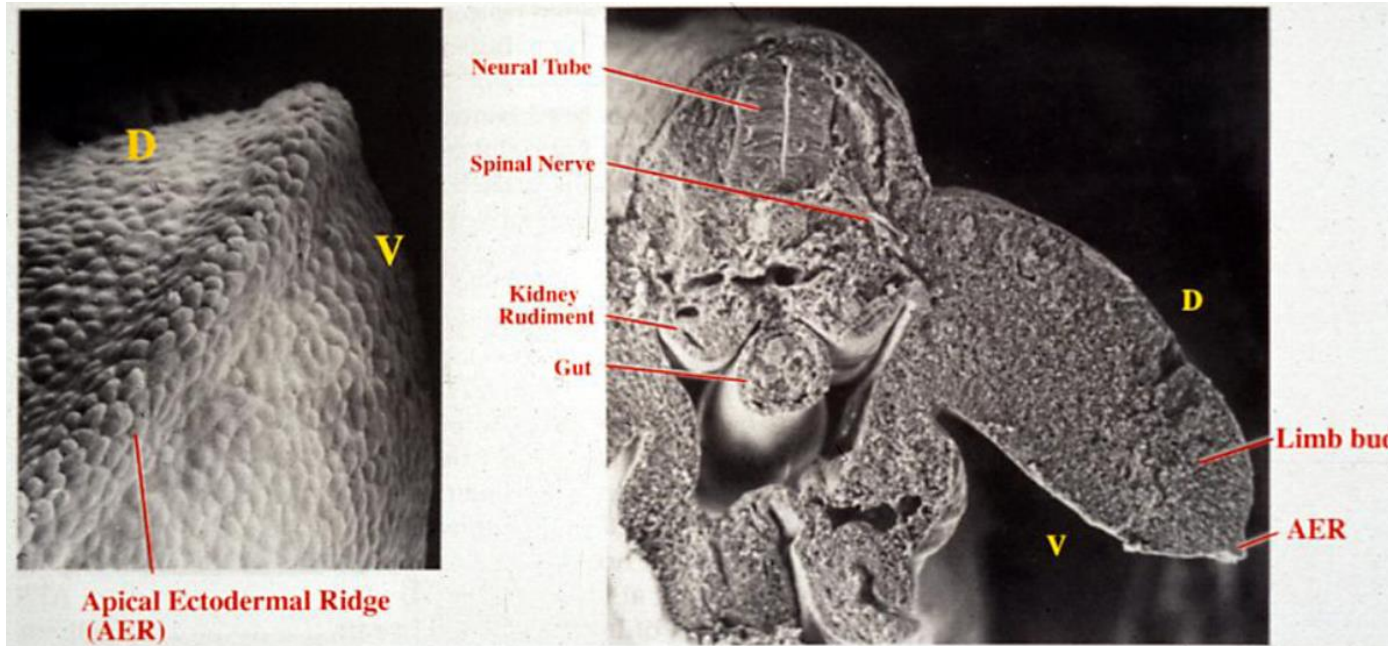
ADAM.



Drugs as Teratogens

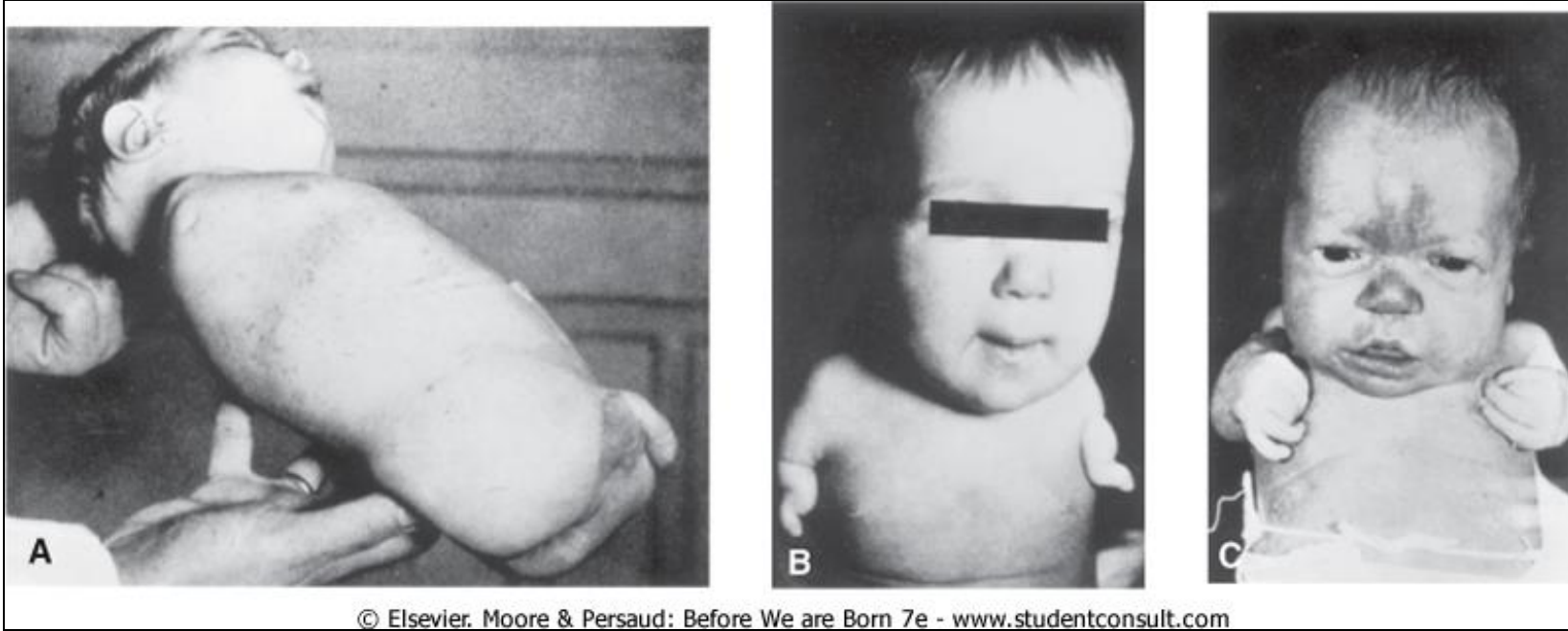
- Thalidomide
- Retinoic acid and its derivatives
- Tetracycline
- Anticonvulsants: Lithium, phenytoin
- Alcohol

Thalidomide and Limb Development



- Thalidomide inhibits Fibroblast growth factors *FGFs* from the apical ectodermal ridge (AER) of the limbs which leads to the proximal limb tissue not developing, while the distal limb tissue continues to differentiate.
- Inhibits angiogenesis

Amelia & Meromelia(phocomelia)



- Amelia
 - Complete absence of limbs
 - Suppression of limb bud development in the 4th week
 - Thalidomide induced Amelia is more frequent in the lower limb
- Meromelia (Phocomelia)
 - Partial absence of limbs
 - Disturbance of limb development during 5th week



Tetracyclines in Pregnancy

- When used during tooth development (second half of pregnancy) tetracyclines may cause permanent yellow-gray-brown discoloration of the teeth and enamel hypoplasia. The use of tetracycline during pregnancy is generally not recommended, especially during the last half of pregnancy.

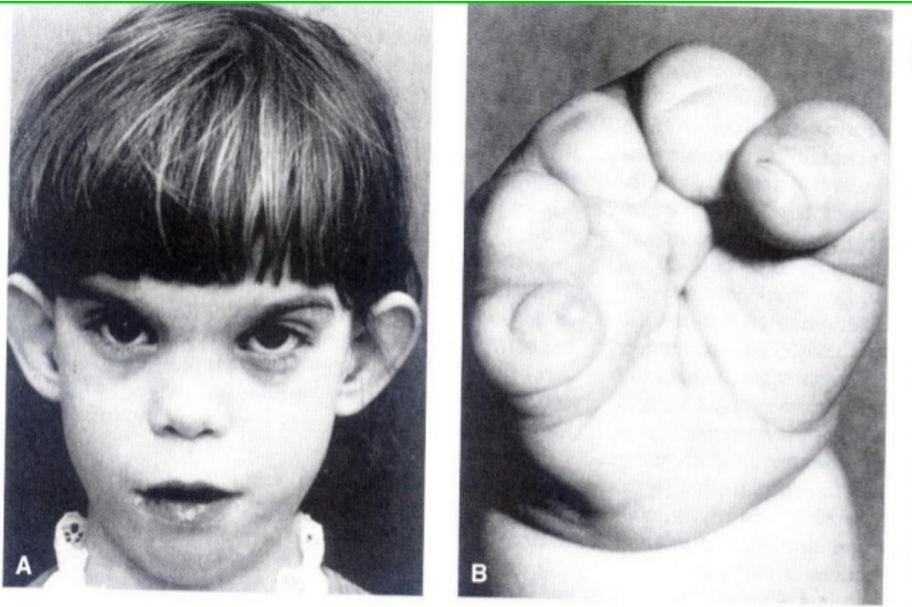


Drugs: Isotretinoin (Retinoic acid) in Pregnancy

- Used for treatment of severe acne
- Severely teratogenic even at low doses.
- Critical period of exposure – 3rd to 5th week
- High doses of vitamin A during pregnancy can produce the same effects.
- **Defects:**
 - spontaneous abortion
 - neuropsychological impairment
 - premature birth
 - neonatal death
- Prevention: **Mandatory contraception**



Anticonvulsive drugs



- **Phenytoin(Dilantin)** affects 5-10% of the embryos based on their genotype
- Fetal hydantoin syndrome.
 - Mental deficiency, IUGR, ptosis hernias, depressed nasal bridge, microcephaly, epicanthal folds and hypoplasia of nails and distal phalanges



- **Valproic acid**
- Drug used in treatment of epilepsy, bipolar disorder and migraines
- Associated defects:
 - Spina bifida, ASD, VSD, Cleft palate, Hypospadias, Polydactyly, Craniosynostosis.
 - Fetal valproate syndrome (rare)
 - Increased risk of autism

Warfarin



Fetal Warfarin Syndrome: infant with hypoplastic nose, flat face and low nasal bridge as well as altered calcification (Smith 1982).

- Warfarin (anticoagulant) inhibits vitamin K epoxide reductase and can cross the placental barrier.
- The critical period 6-12 weeks,
- Hypoplasia of the nasal cartilages, stippled (dotted) epiphysis CNS defects and fetal hemorrhage
- Later trimesters: microcephaly, optic atrophy and some mental disorders.

Organic mercury

Minamata Disease dan Congenital MD



(Photo credit: Masazumi Harada)

<https://image.slidesharecdn.com>

- Organic mercury occurs when bacteria interact mercury present in the sediments in water bodies
- accumulates in the tissue of aquatic animals like fish(swordfish, shark, kingfish)
- Neurotoxic in high concentrations causing loss of nerve cells, decreased brain size and problems with neuronal migration.
- Symptoms : spasticity, seizures mental retardation
- Congenital Minamata disease

Recreational Drugs as Teratogens

- Alcohol
- Tobacco
- Cocaine



Alcohol in Pregnancy

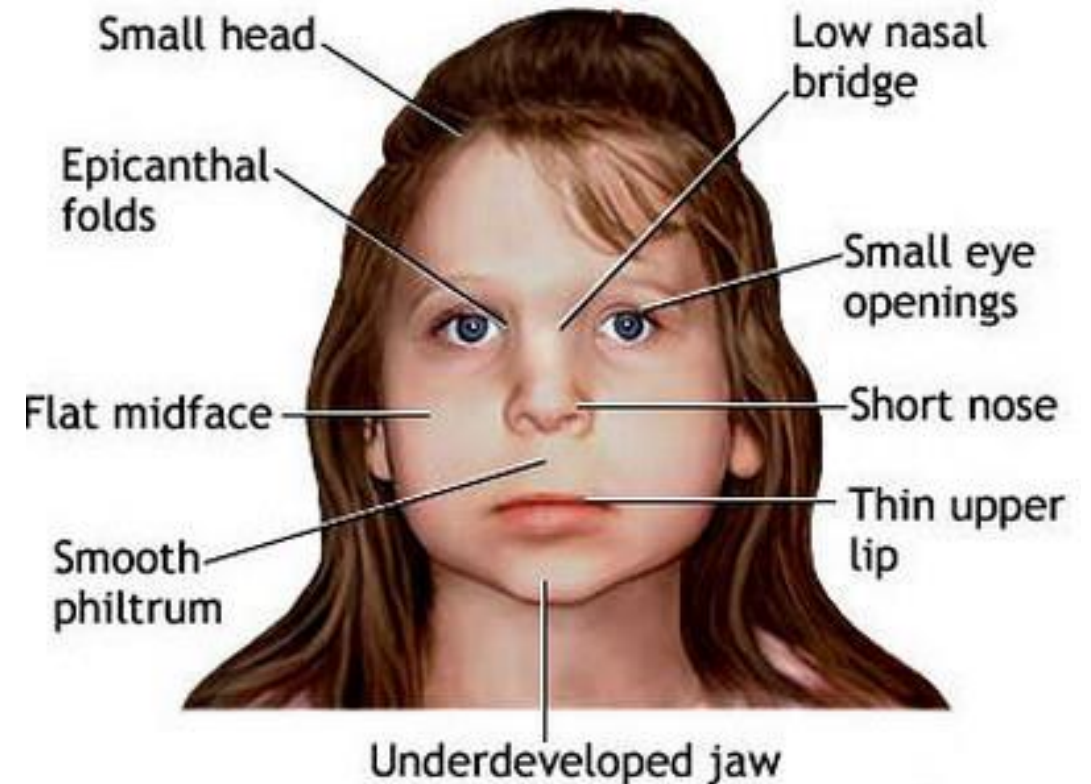
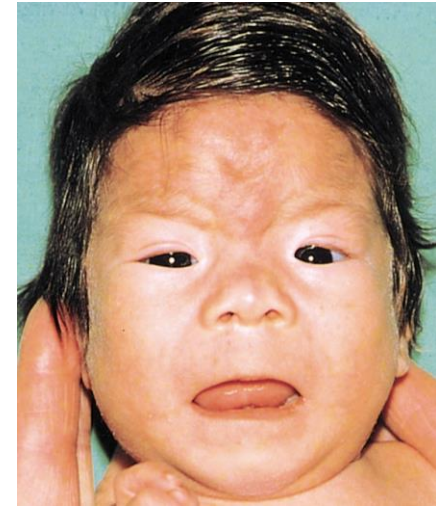
- Alcohol crosses the placenta and remains within the amniotic fluid
 - Risk of birth defects is directly related to the amount and frequency of consumption
 - Disrupts DNA and protein synthesis, inhibits cell migration, disrupts metabolic pathways and the transfer of nutrients across the placenta
 - Spontaneous abortion, preterm delivery and still births

Fetal Alcohol Syndrome (FAS)



Fetal Alcohol Syndrome

- Growth deficiency
- Low IQ (average = 63)
- Mild to moderate microcephaly
- Facial defects: Short nose, smooth philtrum, thin upper lip
- Heart murmur
- Small distal phalanges



Cigarette Smoking in Pregnancy

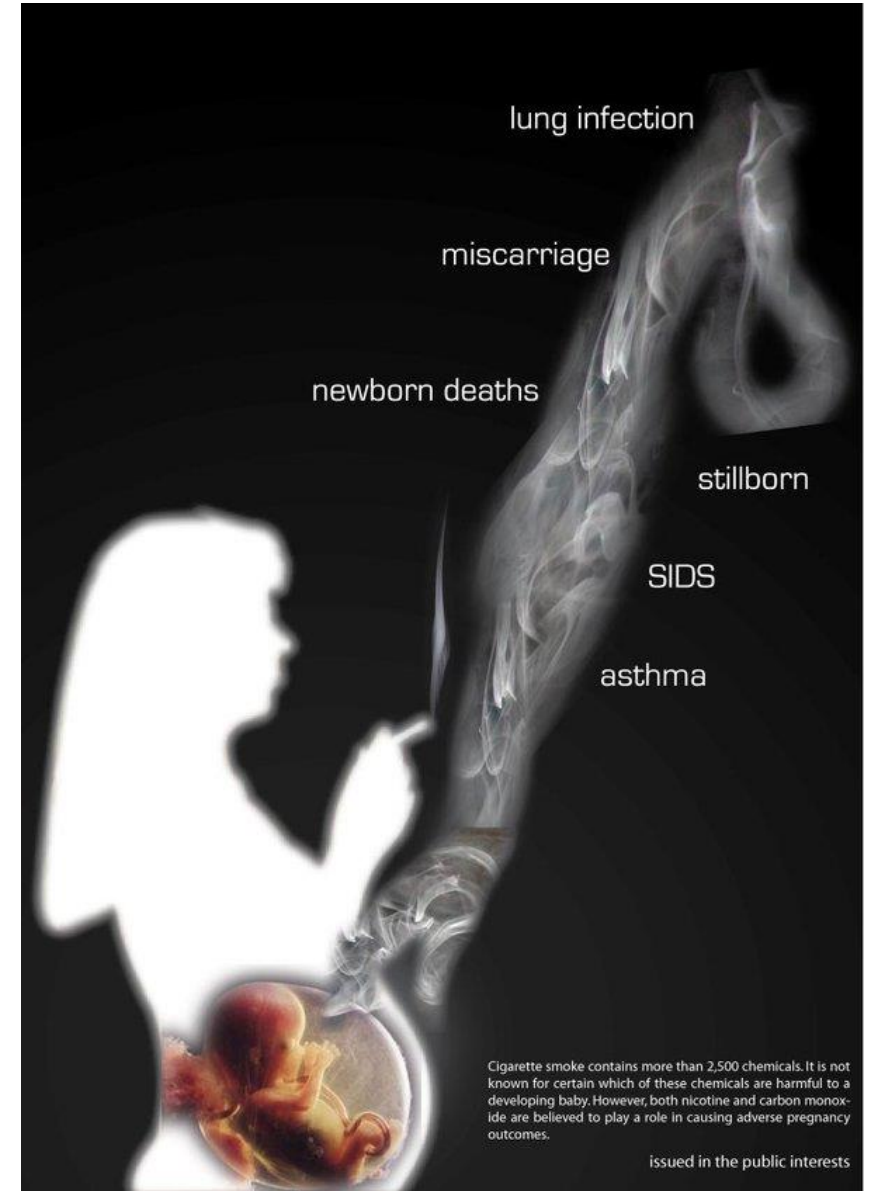
Multiple factors

- Epigenetic changes: DNA methylation
- Nicotine: vasoconstrictor; decreased uterine blood flow
- CO: impaired oxygen delivery fetus

Chronic fetal hypoxia

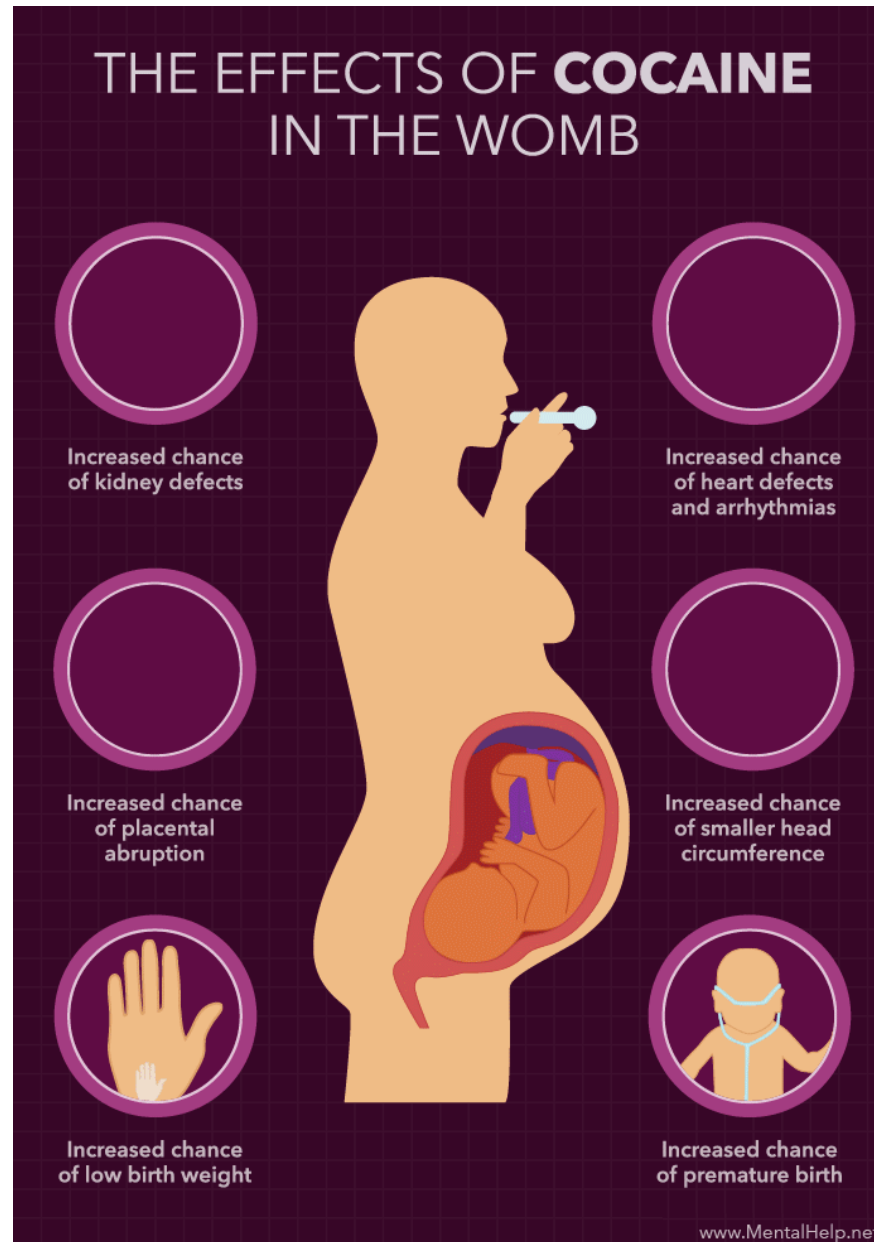
Defects:

- Low birth weight (*Women who stop smoking before pregnancy or 2-3 months into pregnancy, reduce their risk of having a low birth weight baby to that of non smoking women!*)
- Intra uterine growth retardation (IUGR)
- Preterm labor and premature delivery
- Placental defects
- Sudden infant death syndrome (SIDS)
- Conotruncal defects
- Some orofacial (cleft lip and palate)



Cocaine

- Readily crosses the placenta and causes severe vasoconstriction in the fetus
- Can lead to low birth weight
- Probable neurobehavioral disorders
- These babies tend to be easily stimulated and difficult to soothe



<https://i.ytimg.com>

<http://i.dailymail.co.uk>

Cannabis

- Use of cannabis during pregnancy was not associated with increased risk of perinatal mortality or morbidity
- Confounding factors ? *Other drugs, alcohol*
- Cannabis use throughout pregnancy *may* be associated with lower birth weight

Diseases: Maternal Diabetes

Uncontrolled **maternal** diabetes mellitus alters the expression of developmental genes and is associated with the following defects:

- Macrosomia (large baby)
- Impaired growth
- Pulmonary disease (in neonatal period)
- CNS
- Heart: VSD's, transposition of the great arteries, ASD'S and Coarctation of the aorta
- Skeletal
- Metabolic disorders(hypoglycemia, hypocalcemia)



Pavlinkova G et al: Maternal diabetes alters transcriptional programs in the developing embryo. BMC Genomic 10:274 (2009). PubMed 19538749